

A microscopic view of red blood cells, showing their characteristic biconcave disc shape and reddish color. The cells are densely packed in some areas and more sparse in others, creating a textured background. The lighting highlights the surface of the cells, giving them a three-dimensional appearance.

Predicting Sepsis Onset: A Machine Learning-Based Early Warning System for Critical Care

SEPSIS

A Silent Killer

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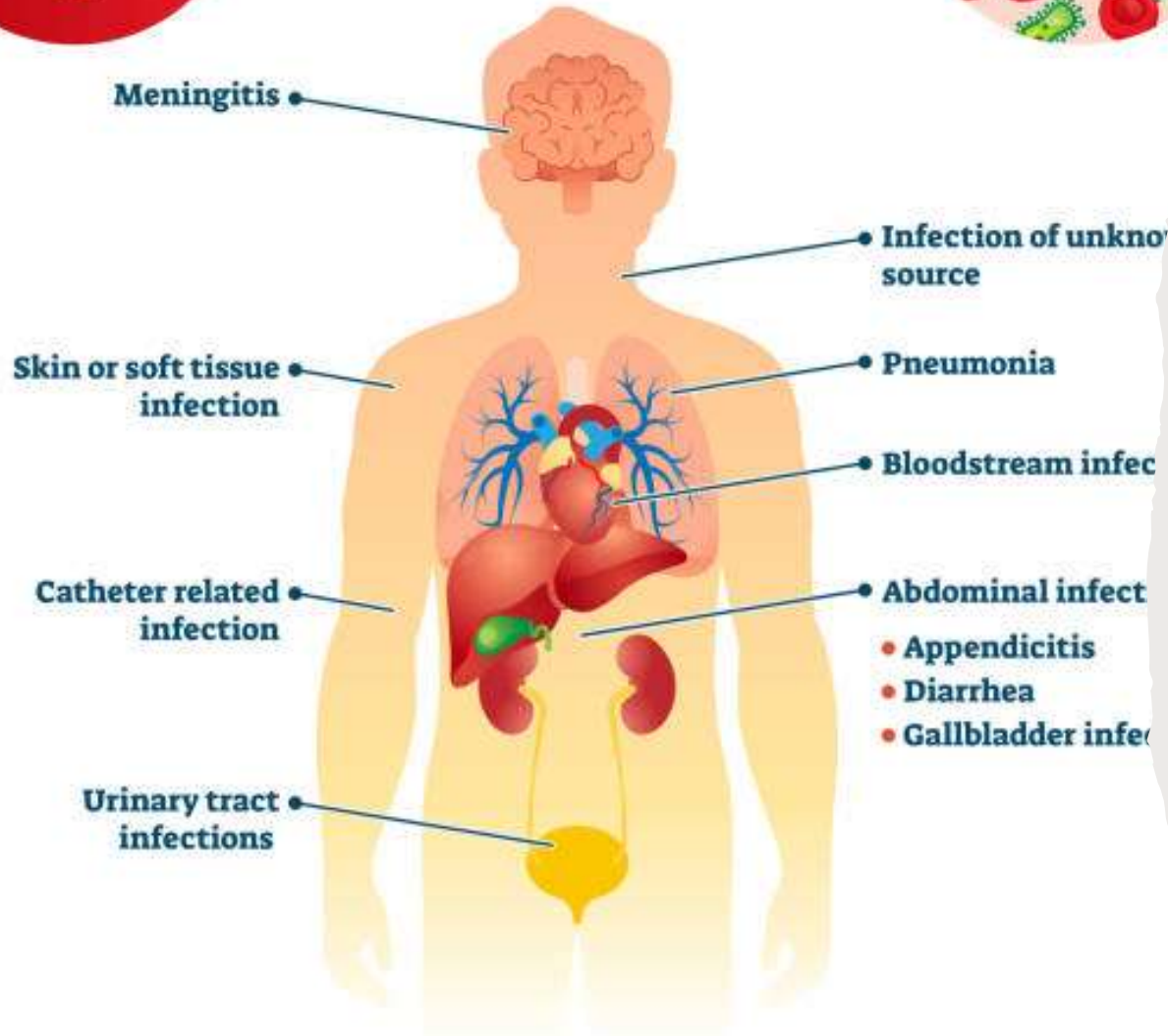
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SEPSIS

Sepsis is a potentially life-threatening condition caused by the **body's response** to an infection



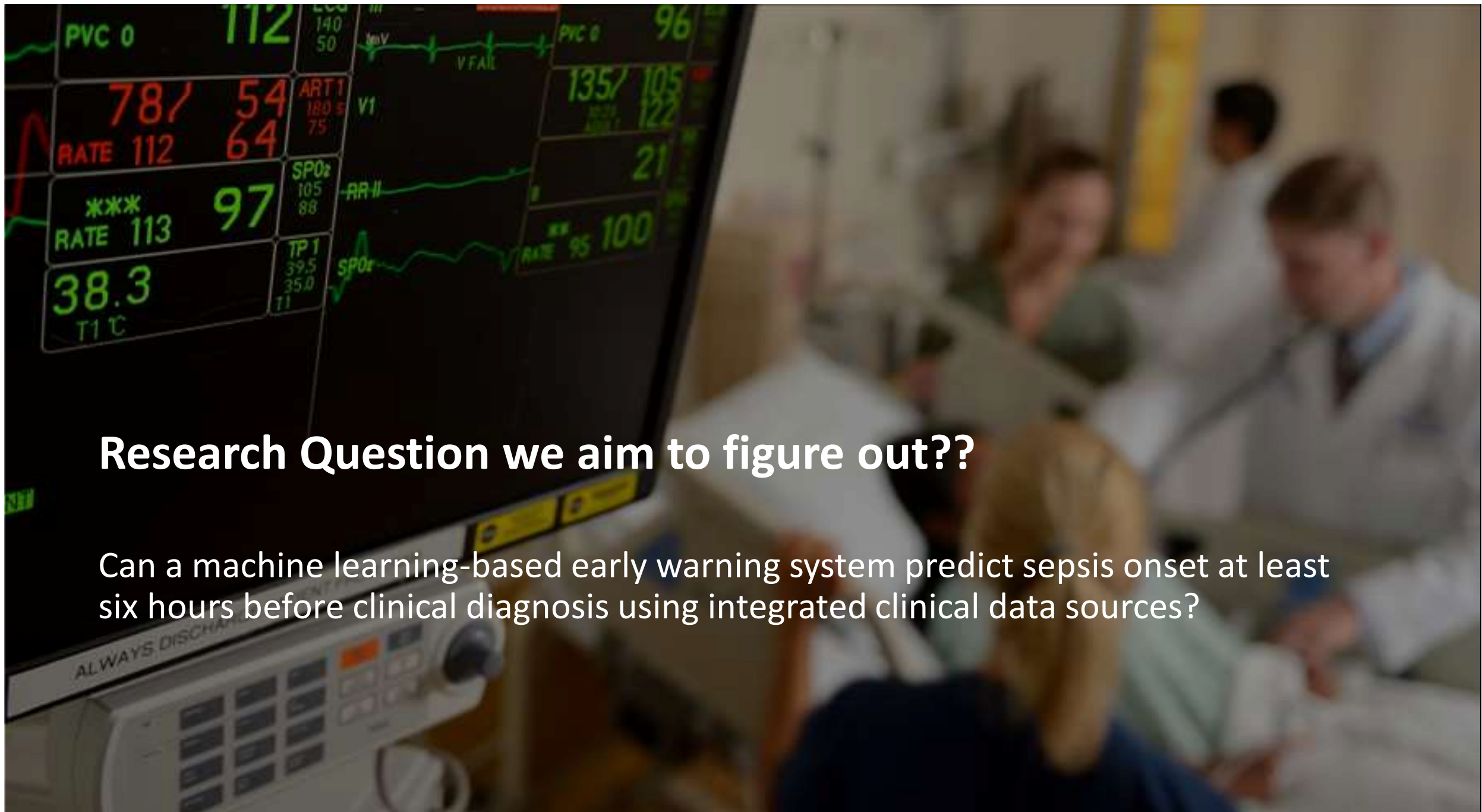
Sepsis is a life-threatening condition where the body's extreme response to an infection causes widespread inflammation, leading to organ failure and death if not treated promptly. It is an emergency because it progresses rapidly, and delayed diagnosis significantly reduces survival chances. Early detection is critical to enable timely intervention, improve patient outcomes, and save lives.

Overview of Relevant Literature

The literature reviewed focuses on the application of **machine learning (ML)** for the early detection of **sepsis**. Early detection is critical, as timely intervention can significantly reduce mortality rates. The studies emphasize the importance of integrating **structured data** example vital signs, lab results and **unstructured data** like clinical notes to improve prediction accuracy. Additionally, the literature highlights the challenges of **imbalanced datasets**, where sepsis cases are rare compared to non-sepsis cases and the need for robust evaluation metrics to ensure the reliability of sepsis prediction models.



Aspect	Islam et al. (2023)	Yadgarov et al. (2024)	Bedoya et al. (2020)
Datasets	ICU EHR (vital signs, lab results, clinical notes)	ICU datasets, PhysioNet 2019 Challenge (time-series data)	ICU datasets (time-series data, vital signs, lab results)
Key techniques	Combining structured and unstructured data for sepsis prediction	Comparison of ML models for sepsis with ICU data	Temporal validation and evaluation metrics for ML models
ML Models	Logistic regression, Random Forests, XGBoost	Logistic regression, Random Forests, XGBoost, Deep learning	Logistic regression, Random Forests, XGBoost, Deep learning
Advantages	More accurate with combined data	Time-series improves prediction, deep learning handles complex patterns	Temporal validation ensures long-term model reliability
disadvantages	Difficult integration of unstructured data	Deep learning needs large data and resources	Temporal validation needs long-term data, feature engineering is time-consuming
Limitations	Model may not generalize across hospitals	Requires continuous high-quality data	Relies on high-quality labeled data



Research Question we aim to figure out??

Can a machine learning-based early warning system predict sepsis onset at least six hours before clinical diagnosis using integrated clinical data sources?

Why Choose This Question?

Sepsis is a life-threatening condition that often gets diagnosed too late, making treatment harder and riskier.

Goal

Predict sepsis 6 hours before it happens using machine learning. This gives doctors enough time to start life-saving treatments like antibiotics and fluids.

Why 6 Hours?

Too early: Might not allow enough time for effective intervention.

Too late: Treatment becomes less effective as sepsis progresses.

Impact:

Reduce deaths

Reduce healthcare costs

Prevent unnecessary treatments.

Opportunity:

With advanced machine learning and quality patient data, we can make a real difference in healthcare.



Methods and algorithms we plan to implement:

1. Data Preprocessing:

- Clean the data by handling missing values and normalizing features e.g, scaling heart rate, blood pressure.
- Use techniques like SMOTE or undersampling to balance the dataset, as sepsis cases are rare compared to non-sepsis cases.

2. Feature selection:

- Analyze the data to identify key patterns and important signs e.g, heart rate, blood pressure, lab results that indicate sepsis risk.
- Create new features based on trends over time e.g., changes in vital signs to improve prediction accuracy.

3. Model Selection and Training:

- Test and compare machine learning models like logistic regression, random forests, and XGBoost to find the best-performing one.
- Fine-tune the models using hyperparameter tuning and cross-validation to ensure they work well on unseen data.

4. Evaluation Metrics:

- We use accuracy, recall, F1-score, and AUROC to evaluate model performance.
- Focus on minimizing false negatives ,missed sepsis cases to ensure early detection.



The dataset consists of **two major categories of features**:

1. Vital Signs:

HR – Heart rate (beats per minute)

O2Sat – Pulse oximetry (%)

Temp – Temperature (°C)

SBP – Systolic blood pressure (mm Hg)

MAP – Mean arterial pressure (mm Hg)

DBP – Diastolic blood pressure (mm Hg) **Resp** –
Respiration rate (breaths per minute)

EtCO2 – End tidal carbon dioxide (mm Hg)

2. Laboratory Values:

BaseExcess – Measure of excess bicarbonate
(mmol/L)

HCO3 – Bicarbonate levels (mmol/L)

FiO2 – Fraction of inspired oxygen (%)

pH – Blood pH level

PaCO2 – Partial pressure of CO2 from arterial blood
(mm Hg)

SaO2 – Oxygen saturation from arterial blood (%)

Datasets Available for the Project:

1. PhysioNet 2019 Dataset (from Kaggle):

- Contains real ICU patient data like heart rate, blood pressure, and lab results recorded over time.
- Designed for early sepsis prediction, rewarding models that predict sepsis 6 hours before diagnosis.

2. MIMIC-III:

- A large, public dataset with ICU patient data, including vital signs, lab results, and doctor's notes.
- Widely used for sepsis prediction and critical care research.

Why We Chose PhysioNet 2019 Dataset: [Prediction of Sepsis](#)

We chose the PhysioNet 2019 Dataset because it's specifically designed for early sepsis prediction, includes real patient data, and focuses on predicting sepsis 6 hours before diagnosis, which matches our project goal. It's also publicly available and has been used in previous research, making it reliable and easy to work with.



Expected Contribution:

To develop a machine learning-based early warning system that predicts sepsis 6 hours before clinical diagnosis. By integrating vital signs, lab results, and clinical notes, we aim to provide healthcare providers with early, actionable alerts, improving patient outcomes and reducing unnecessary treatments.



The image features a scenic mountain landscape with a central graphic element. A large, vibrant purple-to-pink gradient diamond is positioned in the center, containing the words "THANK YOU" in a bold, white, sans-serif font. The background is a high-angle view of a mountain range, showing a dirt path on a grassy slope in the foreground, distant peaks under a cloudy sky, and a valley with a lake. The entire scene is framed by a dark, repeating geometric pattern of rounded hexagons. Faint white concentric circles are also visible behind the central diamond.

**THANK
YOU**